



FARM MANAGEMENT SYSTEM

JAWAHAR SCIENCE COLLEGE, NEYVELI

DEPARTMENT OF COMPUTER SCIENCE AND APPLICATION

COURSE: BACHELOR OF COMPUTER APPLICATION



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ABSTRACT

ABSTRACT

The **Farm Management System** (FMS) is a Salesforce-based application developed to digitize and manage agricultural operations efficiently. The system provides a centralized platform to store and manage data related to farms, crops, farmers, and buyers.

This application helps farmers and farm managers maintain accurate records, track farming activities, and monitor sales and production. By using Salesforce features such as custom objects, validation rules, relationships, reports, and dashboards, the system ensures organized data management and real-time insights.

The Farm Management System reduces manual paperwork, improves data accuracy, and supports better decision-making through analytics and reports. The project aims to simplify farm management processes and enhance productivity by integrating modern cloud technology into agricultural management.

INTRODUCTION

INTRODUCTION

Agriculture plays a vital role in the economy and food production of many countries. Managing farming activities efficiently is essential for improving productivity and maintaining proper records of crops, farms, farmers, and buyers. Traditional farming management methods mainly depend on manual record-keeping, which can be time-consuming and prone to errors.

The **Farm Management System (FMS)** is a cloud-based application developed using the Salesforce platform to digitize and simplify agricultural management. The system helps store, manage, and analyze farm-related data in an organized way. It allows users to maintain information about farms, crops, farmers, and buyers while providing insights through reports and dashboards.

By using Salesforce features such as custom objects, validation rules, relationships, and analytics tools, the Farm Management System improves efficiency, reduces manual work, and supports better decision-making. The system aims to modernize farm management by integrating technology into agricultural operations.

Project view:

The Farm Management System reduces manual record-keeping, improves data accuracy, and enhances decision-making capabilities. It supports farmers and farm managers in tracking operations efficiently, maintaining transparency, and improving overall agricultural management.

Purpose:

The main purpose of this project is to:

- Digitize and automated agricultural operation
- Track agricultural activities efficiently.
- Provide real-time insights through reports and dashboards.
- Monitor sales, Expenses, and financial transactions.
- Enhance overall productivity and operational efficiency.

REQUIREMENTS ANALYSIS PHASE

Customer Journey Map

Stages:

- 1.Login
- 2.Access Farm Management App
- 3.Enter Farm and Crop data
- 4.Register Farmers and buyer
- 5.Generate Reports
- 6.Analyze Results
- 7.Decision Making

DATA FLOW DIAGRAM

Data flow:

The data flow of the Farm Management System (FMS) follows a structured process within the Salesforce platform:

User → Salesforce UI → Custom Objects (Farm, Crop, Farmer, Buyer) → Validation Rules → Relationships → Reports → Dashboards

Explanation:



User: The user (farmer or farm manager) enters or retrieves data through the system.



Salesforce UI: The Salesforce Lightning User Interface is used to input, view, and manage records.



Custom Objects: Data is stored in custom objects such as Farm, Crop, Farmer, and Buyer.



Validation Rules: The system verifies the entered data to ensure accuracy and completeness.



Relationships: Objects are linked using lookup or master-detail relationships to maintain data integrity.

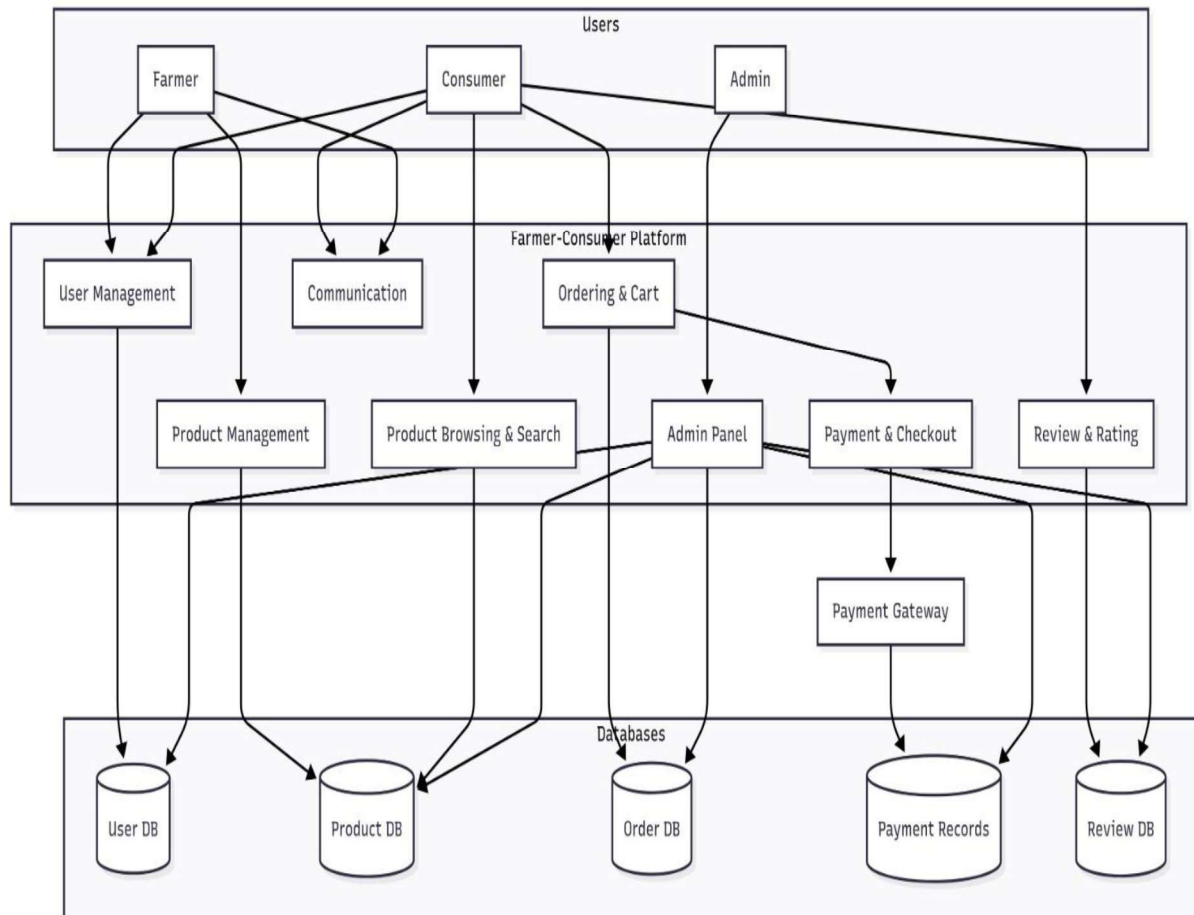


Reports: Data from related objects is analyzed and summarized in reports.



Dashboards: Reports are visualized in dashboards to provide real-time insights and analytics.

ARCHITECTURE DIAGRAM



PROJECT PLANNIGN

5.1 Project Planning

- Created Salesforce Developer Org
- Developed custom objects
- Configured tabs and Lightning App
- Created fields and relationships
- Designed profiles and roles
- Created users and sharing rules
- Developed reports and dashboards
- Conducted testing and reviews

PROJECT DEVELOPMENT PHASE

Project description

Farm Management System (FMS) is developed using the Salesforce platform to provide a comprehensive solution for managing agricultural operations digitally. The system is designed to centralize and organize farm-related data such as farms, crops, farmers, buyers, and sales information.

MILESTONE 1-SALESFORCE

Sales force:

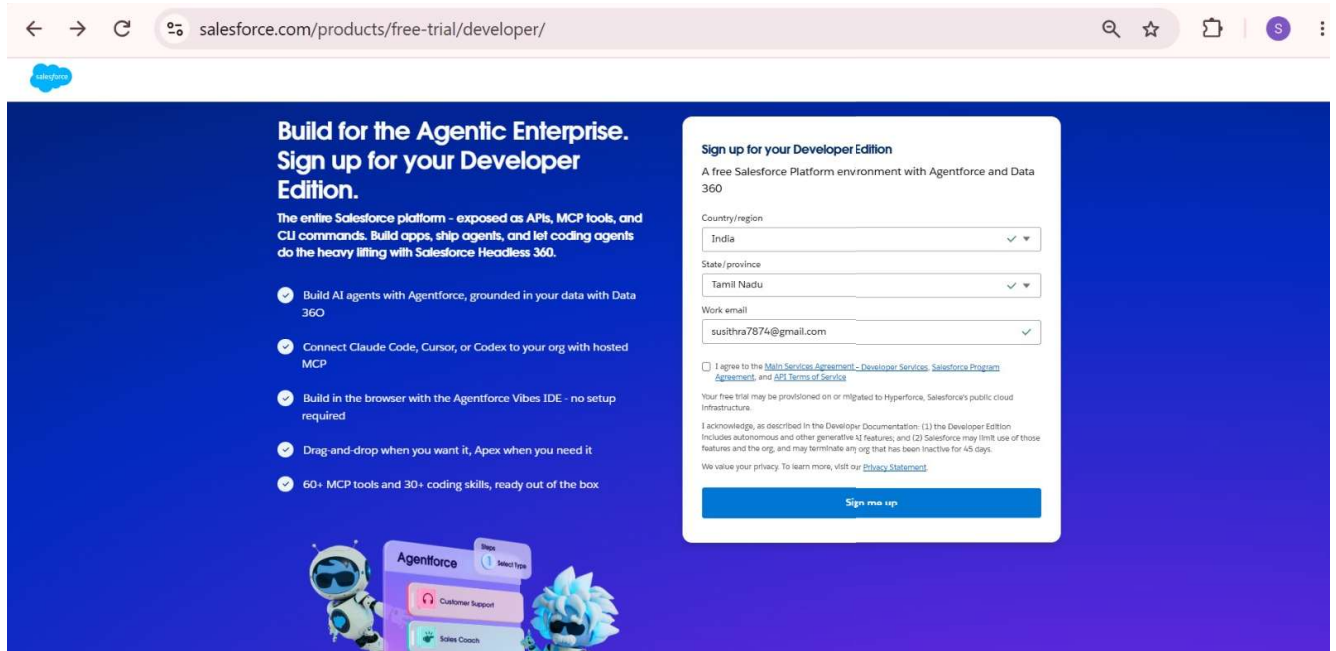
Salesforce is a cloud-based **Customer Relationship Management (CRM)** platform that enables organizations to manage data, automate processes, and generate insights. It provides tools for application development, automation, reporting, and secure cloud storage.

Activity 1:

Creating a Salesforce Developer Org :

A Developer org has all the features and licenses you need to get started with Salesforce.

1. Search Developer.salesforce.com/signup
2. Enter the following details like First name, last name, Email, Role, Company, Country/Region, Postal code, and Username must be unique.
3. Click sign me up, after a few min you will reserve a mail salesforce org and by using the verify account link you can create your new password.
4. Click save.
5. Search login.salesforce.com



← → ↺ salesforce.com/products/free-trial/developer/ 🔍 ☆ 🏠 S ⋮

**Build for the Agentic Enterprise.
Sign up for your Developer Edition.**

The entire Salesforce platform - exposed as APIs, MCP tools, and CLI commands. Build apps, ship agents, and let coding agents do the heavy lifting with Salesforce Headless 360.

- ✓ Build AI agents with Agentforce, grounded in your data with Data 360
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- ✓ Build in the browser with the Agentforce Vibes IDE - no setup required
- ✓ Drag-and-drop when you want it, Apex when you need it
- ✓ 60+ MCP tools and 30+ coding skills, ready out of the box

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A free Salesforce Platform environment with Agentforce and Data 360

Country/region
India ✓

State/province
Tamil Nadu ✓

Work email
susithra7874@gmail.com ✓

☐ I agree to the [Main Services Agreement](#), [Developer Services](#), [Salesforce Program Agreement](#), and [API Terms of Service](#)

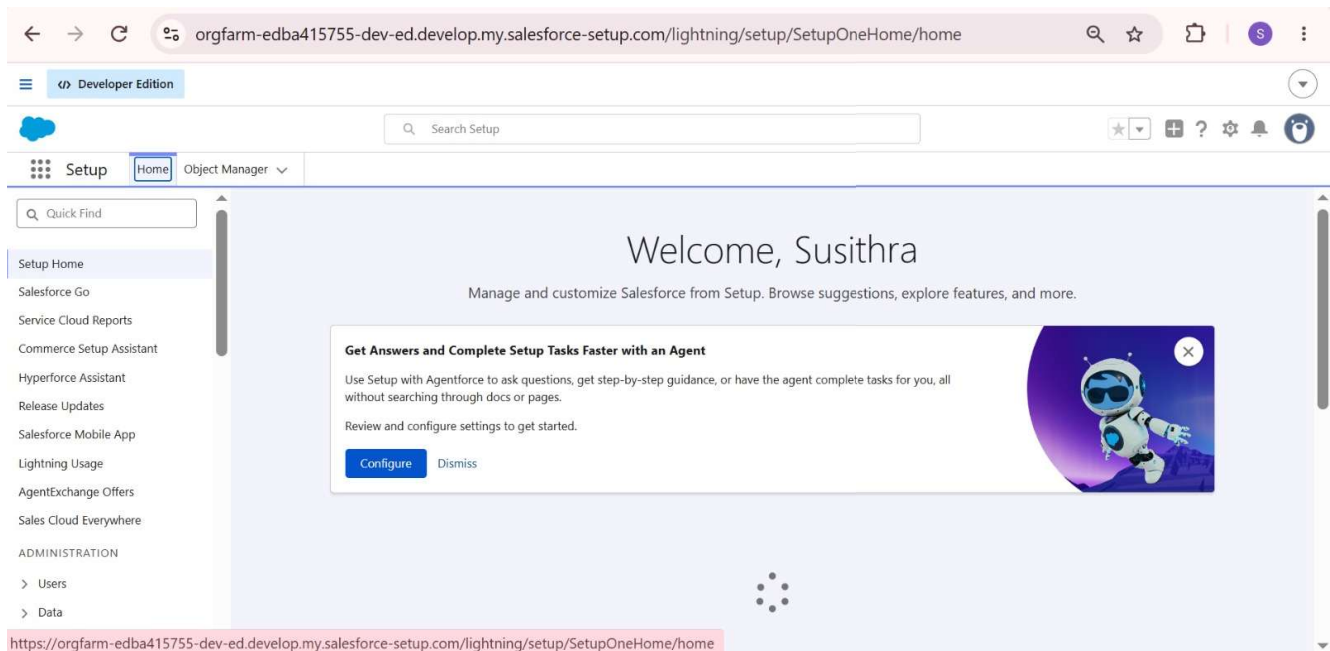
Your free trial may be provisioned on or migrated to Hyperforce, Salesforce's public cloud infrastructure.

I acknowledge, as described in the Developer Documentation: (1) the Developer Edition includes autonomous and other generative AI features, and (2) Salesforce may limit use of those features and the org, and may terminate any org that has been inactive for 45 days.

We value your privacy. To learn more, visit our [Privacy Statement](#).

Sign me up

By using username and password you can into the Salesforce org. The homepage will appear below



MILESTONE 2-OBJECT:

Object

An Object in Salesforce is a database table used to store data in an organized manner. It consists of records (rows) and fields (columns), where each record represents a specific entry and each field stores particular information about that entry.

In This Application We Use 4 Custom Objects:

1. Farm
2. Crop
3. Farmer
4. Buyer

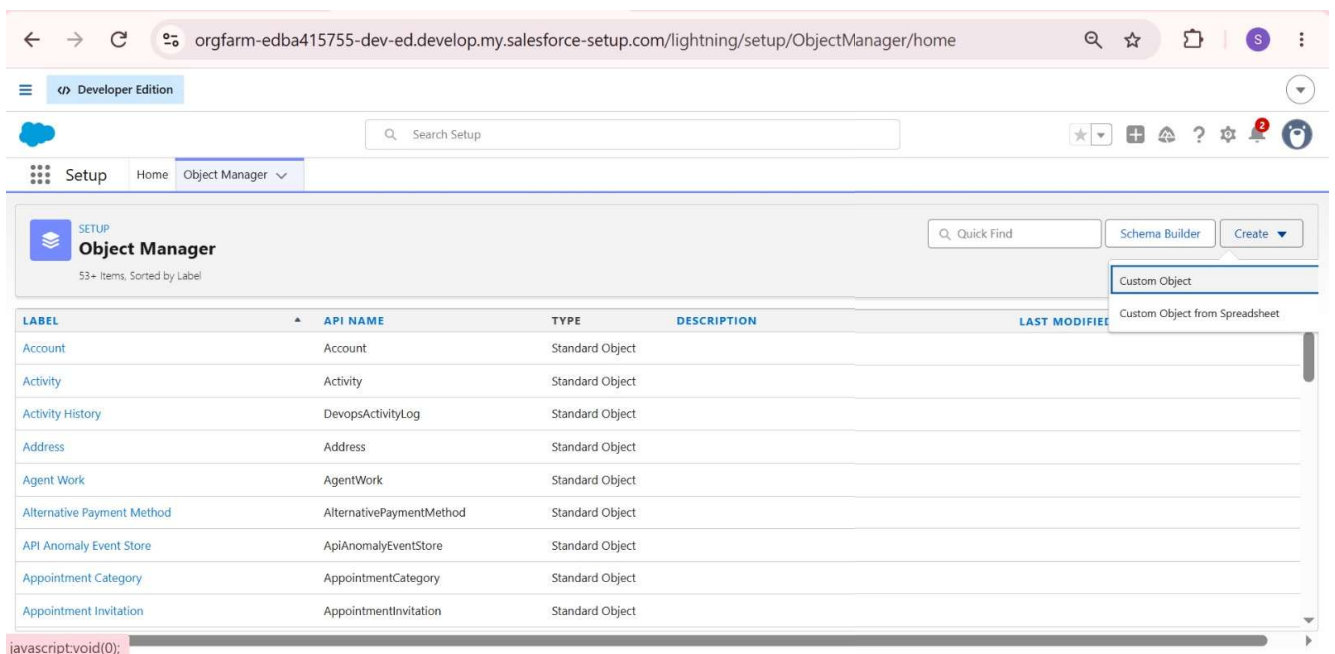
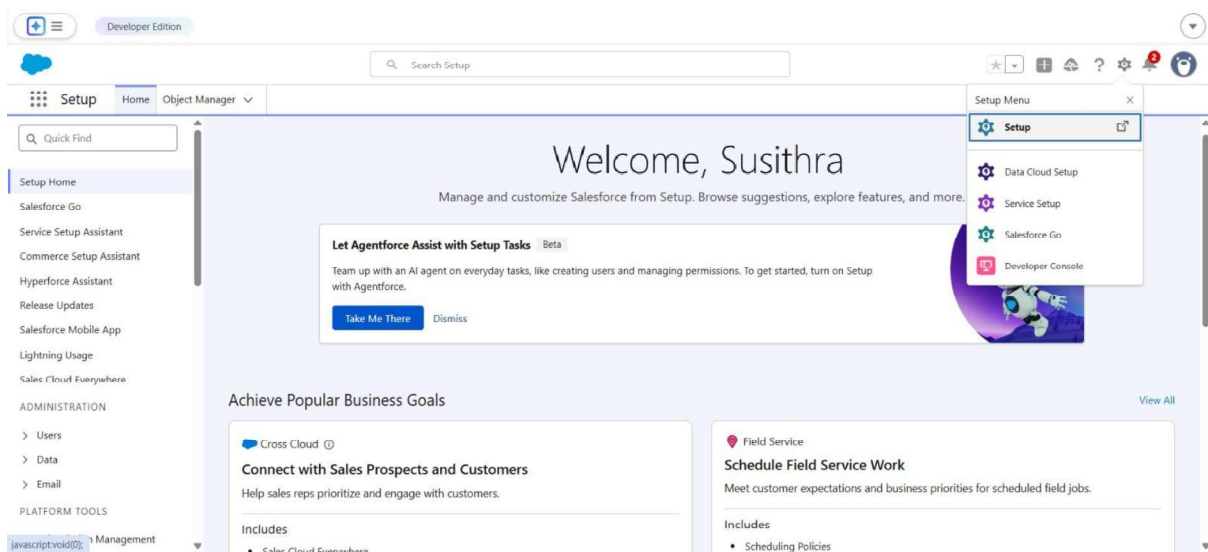
Activity 1:

Create a custom object for Farm:

To create a custom object, follow these steps:

1. From setup click on object manager.

2. Click create, select custom object.

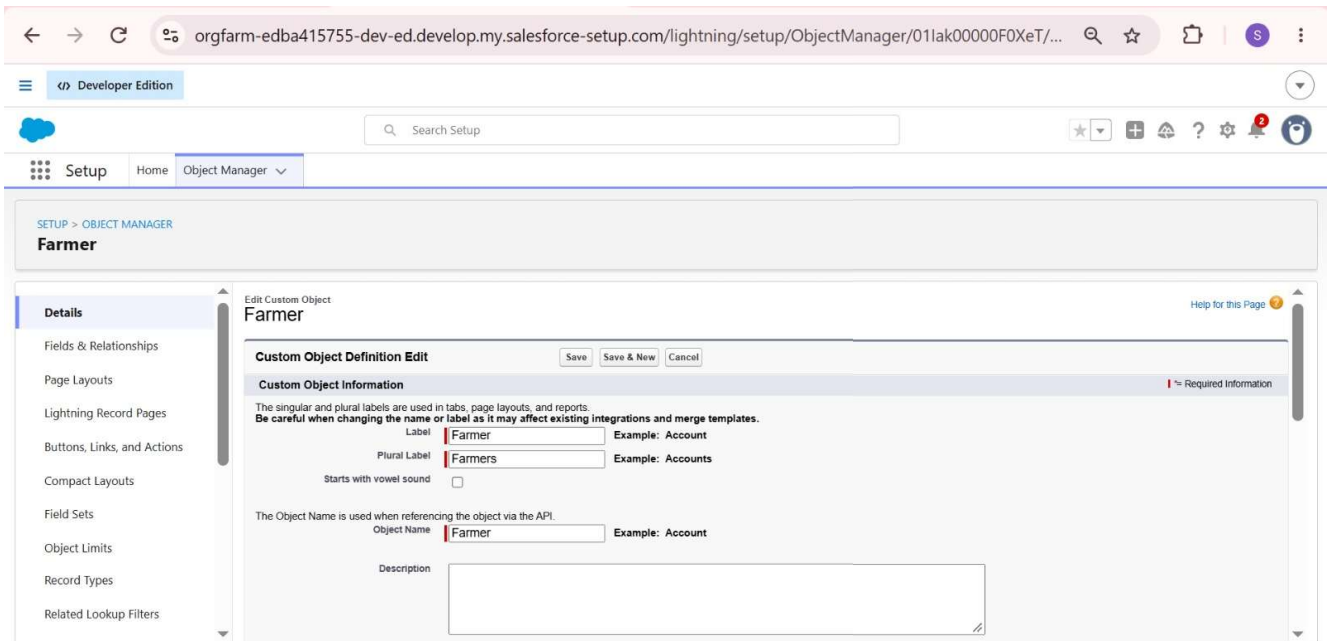


3. Fill in the label as " Farm ".

4. Fill in the plural label as " Farms".

5. Record name: Farm Name"

6. Select the data type as "Text".



7. In the Optional Features section, select Allow Reports and Track Field History.
8. In the Deployment Status section, ensure Deployed is selected.
9. In the Search Status section, select Allow Search.
10. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout.
11. Leave everything else as is and click Save.

Activity 2:

Creation of Crop object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create Drop down click on that and select Custom Object.
 - a. On the Custom Object Definition page, create the object as follows:
 - b. Label: Crop
 - c. Plural Label: Crops
 - d. Record Name: Crop Name
 - e. Select the data type as "Text".
 - f. Check the Allow Reports checkbox
 - g. Check the Allow Search checkbox

h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout if. Click, Save.

Activity 3:

Creation of Farmer object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create

Drop down click on that and select Custom Object.

- a. On the Custom Object Definition page, create the object as follows:
- b. Label: Farmer
- c. Plural Label: Farmers
- d. Record Name: Farmer Name e. Select the data type as "Text".
- f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.
- h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout. i. Click Save.

Activity 4:

Creation of Buyer object:

1. Click on the gear icon and then select Setup.
2. Click on the object manager tab just beside the home tab.
3. After the above steps, have a look on the extreme right you will find a Create

Drop down click on that and select Custom Object.

- a. On the Custom Object Definition page, create the object as follows:
- b. Label: Buyer
- c. Plural Label: Buyers
- d. Record Name: Buyer Name
- e. Select the data type as "Text".
- f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox.
- h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout, Click Save.

MILESTONE-3

Tab: Tab

Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

Activity 1:

Creation of Farm tab:

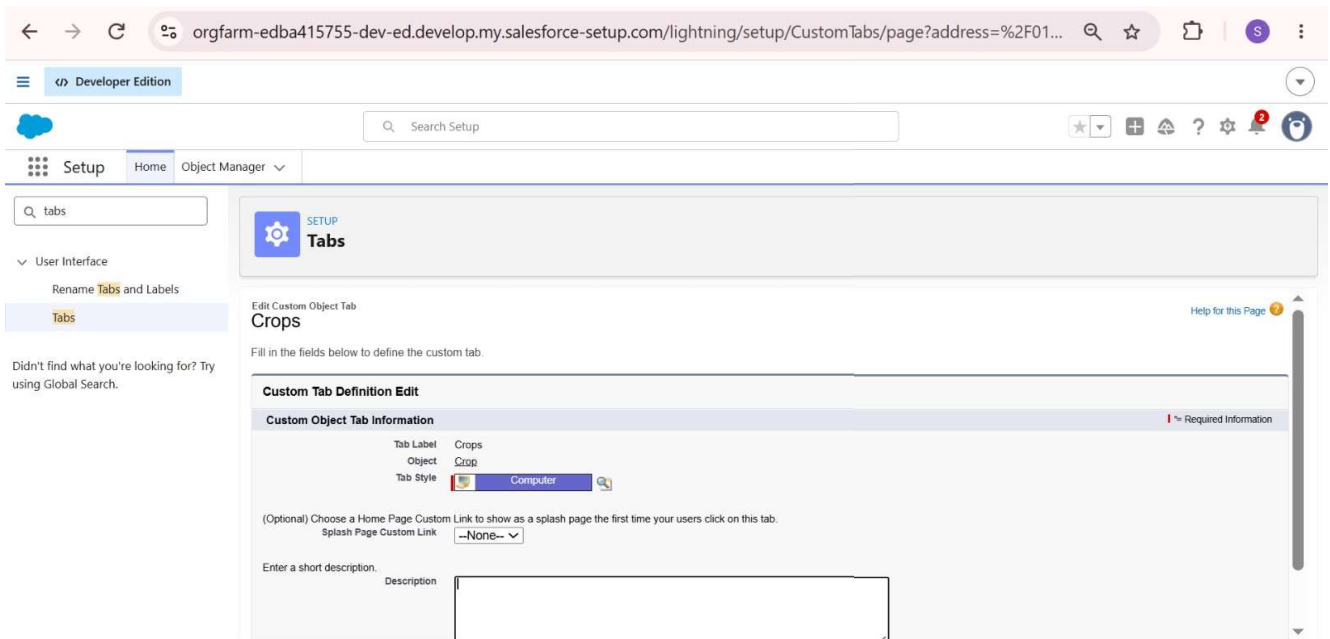
Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.

The screenshot shows the Salesforce Setup interface. The browser address bar displays the URL: `orgfarm-edba415755-dev-ed.develop.my.salesforce-setup.com/lightning/setup/CustomTabs/home`. The page header includes the Salesforce logo, a search bar labeled "Search Setup", and navigation icons. The left sidebar shows the "Setup" menu with "Home" and "Object Manager" options. The "Custom Tabs" section is active, displaying a list of existing custom object tabs. The tabs are: "Buyers" (Big top style), "Crops" (Computer style), "Farmers" (Highway Sign style), and "Farms" (Phone style). Below this list, there is a section for "Web Tabs" which currently shows "No Web Tabs have been defined".

Action	Label	Tab Style	Description
Edit Del	Buyers	Big top	
Edit Del	Crops	Computer	
Edit Del	Farmers	Highway Sign	
Edit Del	Farms	Phone	

2. Under custom object tabs, click New.



2. For Object, select Farm.

5. Leave all defaults as is. Click Next, Next, and Save.

Activity 2:

Creation of Crop tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.

2. Under custom object tabs, click New.

3. For Object, select Crop.

4. For Tab Style, select any icon.

5. Leave all defaults as is. Click Next, Next, and Save.

Activity 3:

Creation of Farmer tab:

Now create a custom tab.

1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.

2. Under custom object tabs, click New.

3. For Object, select Farmer.

4. For Tab Style, select any icon.

5. Leave all defaults as is. Click Next, Next, and Save.

Activity 4:

Creation of Buyer tab:

Now create a custom tab.

1. Click on Home tab, enter Tabs in Quick Find and select Tabs.
2. Under custom object tabs, click New.
3. For Object, select Buyer
4. For Tab Style, select any icon.
5. Leave all defaults as is. Click Next, Next, and Save.

MILESTONE-4/ Lighting app

App:

An App in Salesforce is a collection of related components such as objects, tabs, reports, dashboards, and other features grouped together to perform a specific business function.

Activity 1:

Create the Farm Management system app

orgfarm-edba415755-dev-ed.develop.my.salesforce-setup.com/lightning/setup/NavigationMenus/home

Developer Edition

Search Setup

Setup Home Object Manager

app

Salesforce Mobile App

Data

Mass Transfer Approval Requests

Apps

AgentExchange

Marketplace

App Manager

Builder Central (Beta)

Connected Apps

Connected Apps OAuth Usage

Manage Connected Apps

External Client Apps

External Client App

Lightning Experience App Manager

New Lightning App New External Client App

32 items • Sorted by App Name • Filtered by All app menu items - TabSet Type, App Type

	App Name ↑	Developer Name	Description	Last Modified Date	App Type	Visible...	
1	Agentforce Grid	AIWorkbench	Access Agentforce Grid	9/15/2026, 4:36 AM	Lightning	✓	
2	Agentforce Studio	AgentforceStudio	Agentforce Studio	9/15/2026, 4:36 AM	Lightning	✓	
3	All Tabs	AllTabSet		9/15/2026, 4:27 AM	Classic	✓	
4	Analytics Studio	Insights	Build CRM Analytics dashboards and apps	9/15/2026, 4:27 AM	Classic	✓	
5	App Launcher	AppLauncher	App Launcher tabs	9/15/2026, 4:27 AM	Classic	✓	
6	Approvals	Approvals	Manage approvals and approval flows	9/15/2026, 4:27 AM	Lightning	✓	
7	Automation	FlowsApp	Automate business processes and repetitive tasks.	9/15/2026, 4:34 AM	Lightning	✓	
8	Bolt Solutions	LightningBolt	Discover and manage business solutions designed for your industry.	9/15/2026, 4:27 AM	Lightning	✓	
9	Community	Community	Salesforce CRM Communities	9/15/2026, 4:27 AM	Classic	✓	
10	Content	Content	Salesforce CRM Content	9/15/2026, 4:27 AM	Classic	✓	

- From Setup, enter App Manager in the Quick Find and select App Manager.
- Click New Lightning App. Farm Management System as the App Name, then click Next
- Under App Options, leave the default selections and click Next.
- Under Utility Items, leave as is and click Next.

- From Available Items, select Farm, Crop, Farmer, Buyer

Reports, and Dashboards and move them to Selected Items. Click Next.

- From Available

Profiles, select System Administrator and move it to Selected Profiles. Click Save & Finish.

- To verify your changes, click the App Launcher, type Farm Management System and select the Job Application app.

Note:

1. App Launcher-Displays available apps.
2. App Name-Displays the current selected app.

App Details & Branding

Give your Lightning app a name and description. Upload an image and choose the highlight color for its navigation bar.

App Details

* App Name ⓘ

* Developer Name ⓘ

Description ⓘ

App Branding

Image ⓘ



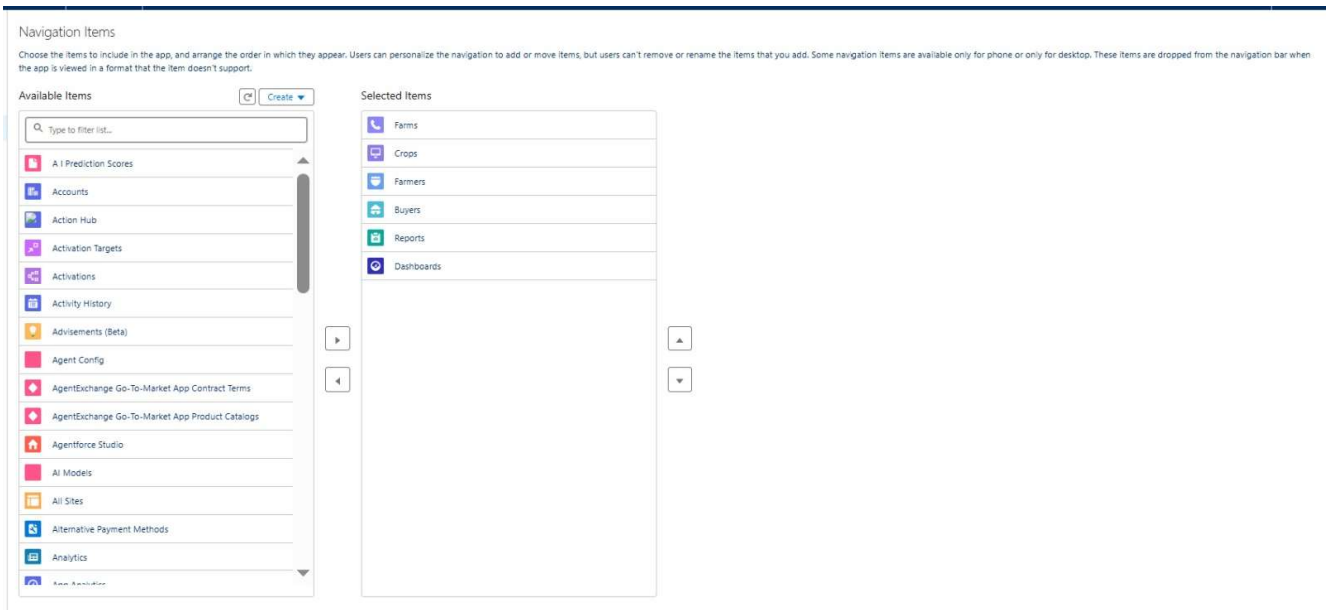
Primary Color Hex Value ⓘ



Org Theme Options

☐ Use the app's image and color instead of the org's custom theme

3.Navigation menu-Displays the tabs available inside the app.



MILESTONR-5

Fields and Relationship:

Fields

Fields in Salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record.

Activity 1:

Creation of fields for the Farm objects:

1. click the gear icon and select Setup. This launches Setup in a new tab.
2. Click the Object Manager tab next to Home.
3. Select Farm.
4. Select Fields & Relationships from the left navigation, and click New

orgfarm-edba415755-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/01lak00000F0VMX...

Developer Edition

Setup Home Object Manager

SETUP > OBJECT MANAGER
Farm

Details

Fields & Relationships
20 Items, Sorted by Field Label

Quick Find New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Actual Yield	Actual_Yield__c	Currency(18, 0)		
Contact Number	Contact_Number__c	Phone		
Created By	CreatedById	Lookup(User)		
Crop Name	Crop_Name__c	Text(100)		
District	District__c	Picklist		
Expected Yield per Acre	Expected_Yield_per_Acre__c	Formula (Currency)		
Farm	Farm__c	Lookup(Farm)		✓
Farm Name	Name	Text(80)		✓

Restriction Rules

5. Choose Data type as a picklist, Click next

Developer Edition

Setup Home Object Manager

SETUP > OBJECT MANAGER
Farm

Details

Fields & Relationships

Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Related Lookup Filters
Search Layouts
List View Button Layout

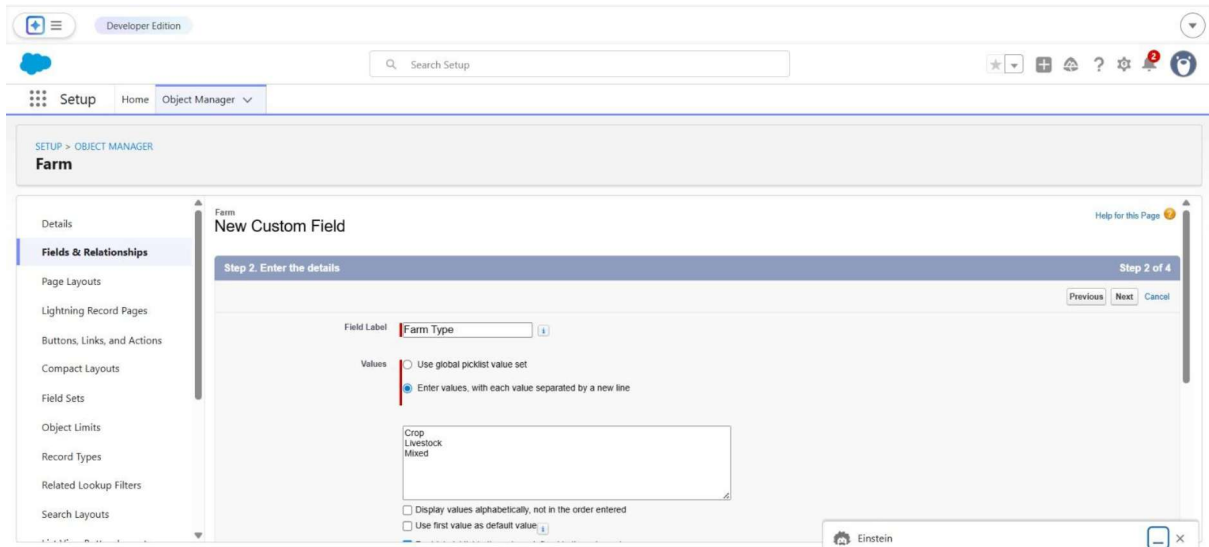
Geolocation
Number
Percent
Phone
☒ Picklist
Picklist (Multi-Select)
Text
Text Area
Text Area (Long)
Text Area (Rich)
Text (Encrypted)
Time
URL

Allows users to define locations. Includes latitude and longitude components, and can be used to calculate distance.
Allows users to enter any number. Leading zeros are removed.
Allows users to enter a percentage number, for example, '10' and automatically adds the percent sign to the number.
Allows users to enter any phone number. Automatically formats it as a phone number.
Allows users to select a value from a list you define.
Allows users to select multiple values from a list you define.
Allows users to enter any combination of letters and numbers.
Allows users to enter up to 255 characters on separate lines.
Allows users to enter up to 131,072 characters on separate lines.
Allows users to enter formatted text, add images and links. Up to 131,072 characters on separate lines.
Allows users to enter any combination of letters and numbers and store them in encrypted form.
Allows users to enter a local time. For example, "2:40 PM", "14:40", "14:40:00", and "14:40:50.600" are all valid times for this field.
Allows users to enter any valid website address. When users click on the field, the URL will open in a separate browser window.

Next Cancel

6. Enter Field Label (Farm Type), and Enter Values as (Crop, Livestock, Mixed)

7. Next, Next Click Save



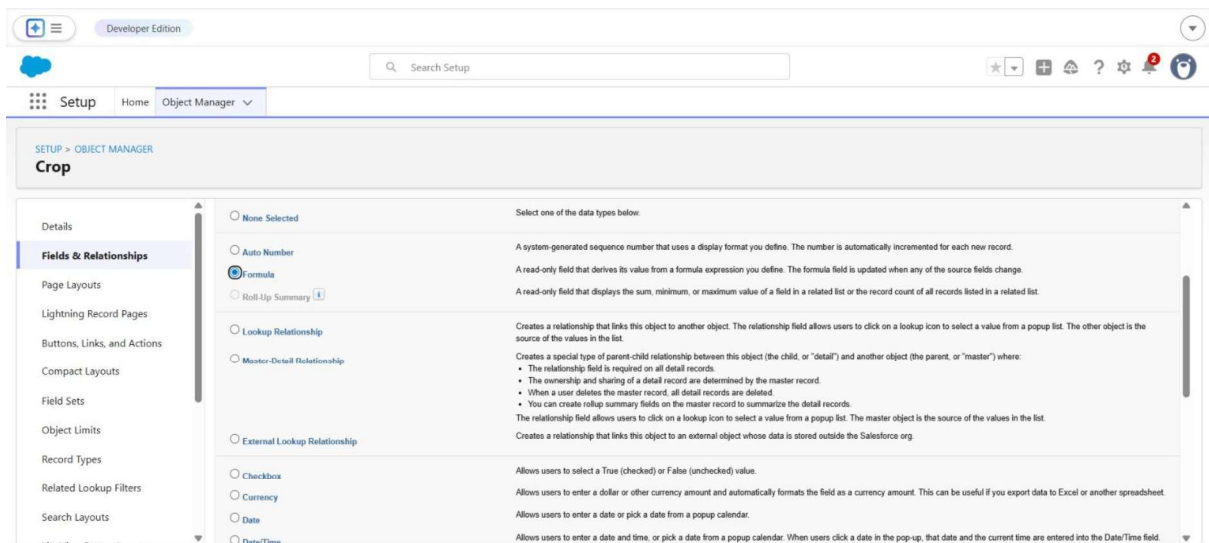
Now let's create the other fields following Activity1 steps 1 to 4 and we must choose the data types of the fields carefully.

- 1.Next we Create another Picklist Field, Enter field label (Soil Quality) enter values (Organic, Cover Crops, Composting)
- 2.Again we Create Picklist field, Field Label (Irrigation type), Enter Values (Sprinkler irrigation, Drip irrigation, localized irrigation)
- 3.Choose data type as a text, enter field label (Location) 4. Choose data type as text, enter field label (size)
- 5.Create a Master Detail Relationship with Farmer Object.

Activity 2:

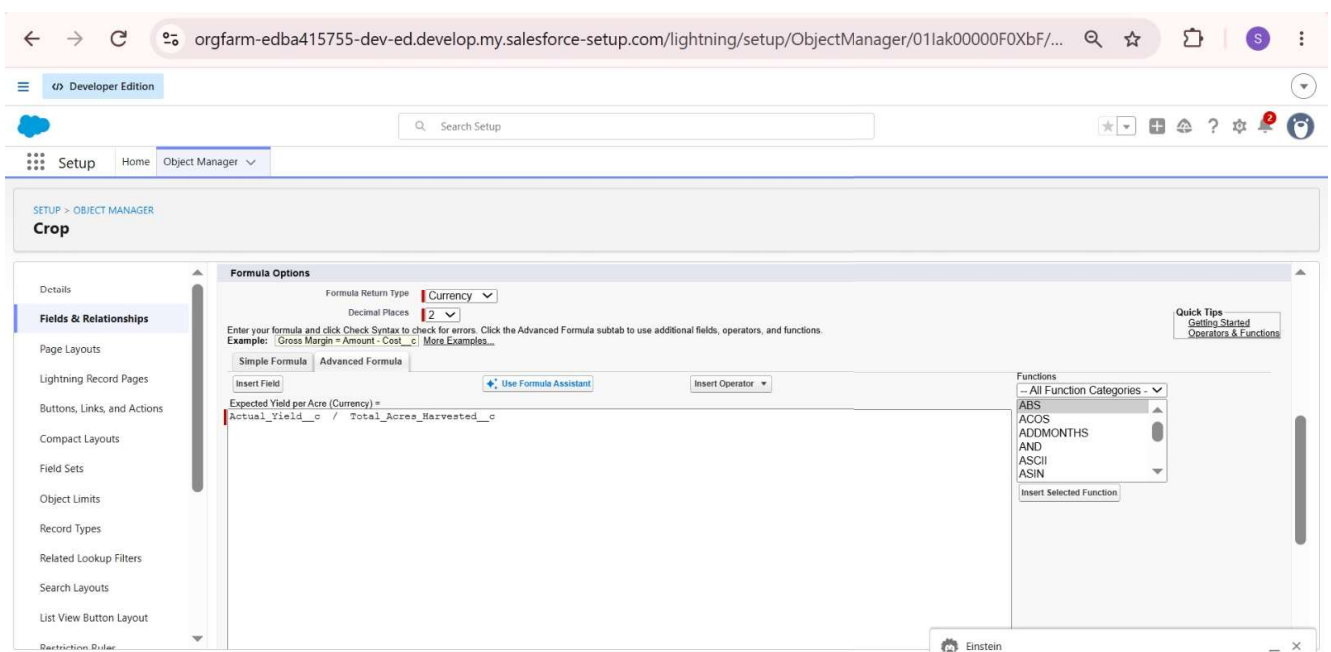
Creation of fields for the Crop objects:

- 1.Choose Data type As a text, Enter Field Label name (Crop Name), Next save. 2.Select Data type as date, Enter Field label (Planting Date), then save.
- 3.Again we select data type as a date, Enter field label (Harvest date).
- 4.Next We Create Lookup relationship on crop object related to Farm Object.
- 5.We select a data type as currency, Enter field Label (Actual Yield), Next Save it.
- 6.Select Data type as a Number, enter Field Label (Total Acres Harvested), Enter Maximum Length (20) and save.
- 7.Now We create Formula Field For That Select data Type As a Formula, Click Next.



8. Enter Field Label (Expected Yield per Acre), Select formula return type as Currency, click Next.

9. Enter a Formula Actual Yield Divide By Total Acre Harvested, then click next and save.



Activity 3:

Creation of fields for the Farmer objects:

1. Select Data type Phone, Enter field label (Contact Number), next Save.
2. Select Data type Picklist, Enter Field Label (State), Enter values (Telangana, Andhra Pradesh, Tamil Nadu) , save.
3. Choose Data type Picklist, Enter Field Label (District), Enter Values of district, next save.
4. Again select Data type as Picklist, Enter Field Label (Village), Enter Values of villages, next save.

Creation of field dependency :

1. Now create Field Dependency For District field, Open Object Manager, Choose Farmer Object, select Fields& Relationships, On Right side We have Field Dependency option, select that option.
2. Click new, Select Controlling Field as State, Dependent Field As District, Click on Continue.
3. Select the appropriate Value in each column by double-clicking them.
4. For Ex. Telangana – Yadadri Bhongir.
5. Click Include Values. And it is also the same for Andhra Pradesh, Tamilnadu with its city.
6. Click Preview, then test the dependency by selecting a different State and viewing the Associate Values available for particular state.

Developer Edition

Search Setup

SetupHomeObject Manager

SETUP > OBJECT MANAGER

Farmer

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Search Layouts

New Field Dependency

Create a dependent relationship that causes the values in a picklist or multi-select picklist to be dynamically filtered based on the value selected by the user in another field.

- The field that drives filtering is called the "controlling field." Standard and custom checkboxes and picklists with at least one and less than 300 values can be controlling fields.
- The field that has its values filtered is called the "dependent field." Custom picklists and multi-select picklists can be dependent fields.

Step 1. Select a controlling field and a dependent field. Click Continue when finished.

Step 2. On the following page, edit the filter rules that control the values that appear in the dependent field for each value in the controlling field.

Controlling Field

Dependent Field

ContinueCancel

ContinueCancel

Developer Edition

Search Setup

SetupHomeObject Manager

SETUP

Quick Find

Setup Home

Salesforce Go

Service Setup Assistant

Commerce Setup Assistant

Hyperforce Assistant

Release Updates

Salesforce Mobile App

Lightning Usage

Sales Cloud Everywhere

ADMINISTRATION

> Users

> Data

> Email

PLATFORM TOOLS

Click button to include or exclude selected values from the dependent picklist:

Include ValuesExclude Values

Showing Columns: 1 - 3 (of 3) < Previous | Next > View All Go to

State:	Telangana	Andhra Pradesh	Tamil Nadu
District:	Hyderabad	Hyderabad	Hyderabad
	Secunderabad	Secunderabad	Secunderabad
	Bhongir	Bhongir	Bhongir
	Nalgonda	Nalgonda	Nalgonda
	Vishakhapatnam	Vishakhapatnam	Vishakhapatnam
	Vijayawada	Vijayawada	Vijayawada
	Nellore	Nellore	Nellore
	Amaravati	Amaravati	Amaravati
	Chennai	Chennai	Chennai
	Coimbatore	Coimbatore	Coimbatore
	Kanchipuram	Kanchipuram	Kanchipuram
	Kanniyakumari	Kanniyakumari	Kanniyakumari

Showing Columns: 1 - 3 (of 3) < Previous | Next > View All

Click button to include or exclude selected values from the dependent picklist:

Include ValuesExclude Values

SaveCancelPreview

https://orgfarm-623640fb05-dev-ed.develop.my.salesforce.com/lightning/setup/null/page?address=%2Fp%2Fdependency%2FeditDependencyUI%2Fe%3FretURL%3D%252Fsetup%252FObjectManager%252F01lg.000004edwP%252FFieldsAndRelationships%252FviewDepend...

7. Click close to close the preview window, click save.

Activity 4:

Creation of fields for the Buyer objects:

1. Select Data type as a Phone, Enter a Field Label (Contact), Save.
2. Select Data type as an Email, Enter Field Label (Email), Next save.
3. Select Data type Text area, Enter Field Label (Address), Save it.
4. Select Data type Picklist, Field Label (Buyer Type), Enter values (Wholesale, Retail), Click Save.

MILESTONE-6/ profile

A profile is a group/collection of settings and permissions that define what a user can do in salesforce. A profile controls “Object permissions, Field permissions, User permissions, Tab

settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types, Login hours & Login IP ranges.

A profile can be assigned to many users, but user can be assigned single profile at a time.

Activity 1:

Create a custom profile:

1. From setup, enter profiles in Quick Find box
2. Select profiles (Standard user).
3. Click clone.
4. For Profile, enter Sales Department.

← → ↻ orgfarm-edba415755-dev-ed.develop.my.salesforce-setup.com/lightning/setup/EnhancedProfiles/home 🔍 ☆ 📄 | S ⋮

Developer Edition

Setup Home Object Manager

Q pro

Hyperforce Assistant

- Users
 - Profiles**
- Data
 - Mass Transfer Approval Requests
- Feature Settings
 - Approval Settings
 - Data.com
 - Prospector Preferences
 - Prospector Users
- Functions
- Marketing
 - Lead Processes
- Sales
 - Products
- Asset Settings

SETUP Profiles

Profiles

All Profiles Edit Delete Create New View

New Profile

Action	Profile Name *	User License	Custom
☐ Edit Clone	Work.com Only User	Work.com Only	☐
☐ Edit Clone	System Administrator	Salesforce	☐
☐ Edit Clone	Standard User	Salesforce	☐
☐ Edit Clone	Standard Platform User	Salesforce Platform	☐
☐ Edit Clone	Solution Manager	Salesforce	☐
☐ Edit Clone	Silver Partner User	Silver Partner	☐
☐ Edit Del ...	Salesforce API Only System Integrations	Salesforce Integration	☒
☐ Edit Del ...	Sales Department	Salesforce	☒
☐ Edit Del ...	Read Only	Salesforce	☒
☐ Edit Clone	Partner Community User	Partner Community	☐
☐ Edit Clone	Partner Community Login User	Partner Community Login	☐
☐ Edit Clone	Partner App Subscription User	Partner App Subscription	☐

1-25 of 46 0 Selected Previous Next Page 1 of 2

← → ↻ orgfarm-edba415755-dev-ed.develop.my.salesforce-setup.com/lightning/setup/EnhancedProfiles/page?address=%... 🔍 ☆ 📄 | S ⋮

Developer Edition

Setup Home Object Manager

Q pro

Hyperforce Assistant

- Users
 - Profiles**
- Data
 - Mass Transfer Approval Requests
- Feature Settings
 - Approval Settings
 - Data.com
 - Prospector Preferences
 - Prospector Users
- Functions
- Marketing
 - Lead Processes
- Sales
 - Products
- Asset Settings

SETUP Profiles

Clone Profile

Enter the name of the new profile.

You must select an existing profile to clone from. Required Information

Existing Profile	Standard User
User License	Salesforce
Profile Name	Farm manager

Save Cancel

5.click save.

Activity 2:

Create a custom profile-2:

Create a profile with the profile name as “Farm Manager”.

6. From setup, enter profiles in Quick Find box
7. Select profiles (Standard user).
8. Click clone.

MILESTONE-7/Role

In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

Activity-1

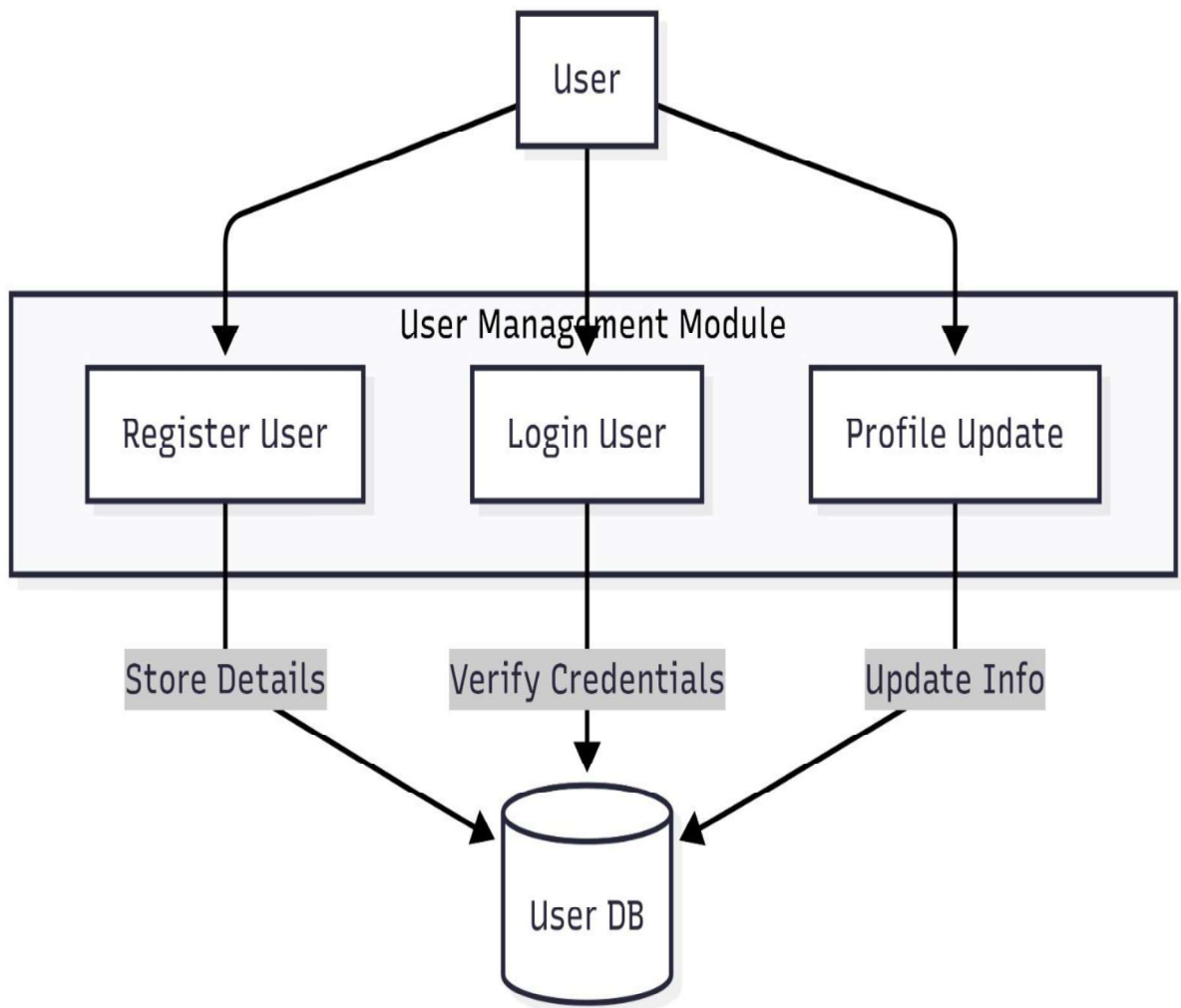
Creation of role

1. From the Quick find box search for the role and click on the roles option
2. Select the setup role option

The screenshot displays the Salesforce Setup interface for the 'Roles' section. The left sidebar shows the navigation menu with 'Roles' selected. The main content area is titled 'Understanding Roles' and includes a 'Sample Role Hierarchy' diagram. The diagram illustrates a hierarchy starting with 'Executive Staff' (CEO, President, CFO, VP, Sales) at the top. Below this are three 'Sales Director' roles: 'Western Sales Director', 'Eastern Sales Director', and 'International Sales Director'. Each director role has associated 'Sales Rep' roles: 'CA Sales Rep' and 'OR Sales Rep' for Western, 'NY Sales Rep' and 'MA Sales Rep' for Eastern, and 'Asian Sales Rep' and 'European Sales Rep' for International. The diagram also includes text boxes explaining permissions for each level, such as 'View & edit data, roll up forecasts, & generate reports for all users below' for the Executive Staff and 'View & edit data, roll up forecasts, & generate reports only for own data' for the Sales Reps. At the bottom of the diagram, there are buttons for 'Set Up Roles' and 'Don't show this page again'.

3. Below the CEO click on add role and enter the label name as a “Farm Manager” and role name will be Automatically populated and click on save.

USERS



MILESTONE-8/USER

User

A user is anyone who logs into Salesforce. Users are employees at your company, such as sales reps, managers, and IT specialists, who need access to the company's records. Every user in

Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and record the user can access.

Activity 1:

To Create a user:

- 1.From Setup, enter Users in the Quick Find box, then select Users.
- 2.Click new user

The screenshot shows the Salesforce Setup page for Users. The left sidebar contains a navigation menu with options like Analytics Groups, Permission Set Groups, Permission Sets, Profiles, Public Groups, Queues, Roles, User Management Settings, and Users. The main content area is titled 'All Users' and includes a search bar and a 'New User' button. Below the button is a table of existing users.

Action	Full Name	Alias	Username	Role	Active	Profile
☐ Edit	Y. Susithra	sus	susithra7874.9e7b7047a20@agentforce.com		✓	System Administrator
☐ Edit	User_Security	sec	insightsecurity@00da00001gs2xdea1.com		✓	Analytics Cloud Security User
☐ Edit	User_Integration	integ	integration@00da00001gs2xdea1.com		✓	Analytics Cloud Integration User
☐ Edit	Manager_Farm	fmna	farm@gmail.com	Farm Manager	✓	Standard Platform User
☐ Edit	Manager_Farm	fmna	dj@kahnew232@gmail.com	Farm Manager	✓	Farm manager
☐ Edit	EPIC_OrgFarm	OEPI	epic.c8dc88c91ae@orgfarm.salesforce.com		✓	System Administrator
☐ Edit	destaf_Susithra	sdest	sdest@gmail.com	Sales Department	✓	Standard User
☐ Edit	Chatter_Expert	Chatter	chatter.00da00001gs2xdea1.kabjoo0gr7c6@chatter.salesforce.com		✓	Chatter Free User

3. Enter First name as Farm and last name as Manager.
4. Enter the user's name and email address and a unique username in the form of an email address. By default, the username is the same as the email address.
5. Select the role of Farm Manager.
6. Select user License as Standard Platform User.
7. Select profile (Farm Manager).
8. Click save.

Activity 2:

Create a user with a username as “Sales Department”, and assign him the sales Manager profile. Follow the steps from above Activity

MILSTONE-9/Sharing Rules

Sharing Rules

Sharing rules help users to share records based on conditions. It is basically created for objects whose organization-wide defaults (OWD) are set to public read-only or private because sharing rules can only extend the access and not restrict it.

Types of sharing rules,

•Owner-based Sharing Rules •Criteria-based Sharing Rules

Activity 1:

Create a sharing rule

1. Goto Sharing Settings, which can be found under the Quick Find section.
2. Scroll down and find the candidate object where a sharing rule needs to be added, and then click on New to create a new sharing rule.
3. Add the label of the sharing rule you want to make.
4. Select your rule type based on the criteria.
5. Select the field can join immediately check field from the Crop object.
6. Select the Crop as equal and value is True.
7. And in selecting the users to share with the section select roles and in that select Farm Manager.
8. And in the section of select the level of access for the users give the access read/write.
9. And save Rule.

Sharing Settings

This page displays your organization's sharing settings. These settings specify the level of access your users have to each others' data. Go to [Background Jobs](#) to monitor the progress of a change to an organization-wide default or a parallel sharing recalculation.

Manage sharing settings for: All Objects

[Disable External Sharing Model](#)

Default Sharing Settings

Object	Default Internal Access	Default External Access	Grant Access Using Hierarchies
Lead	Public Read/Write/Transfer	Private	✓
Account and Contract	Public Read/Write	Private	✓
Contact	Controlled by Parent	Controlled by Parent	✓
Order	Controlled by Parent	Controlled by Parent	✓
Asset	Controlled by Parent	Controlled by Parent	✓
Opportunity	Public Read/Write	Private	✓
Case	Public Read/Write/Transfer	Private	✓
Campaign	Public Full Access	Private	✓
Campaign Member	Controlled by Campaign	Controlled by Campaign	✓

Activity 2:

Create a Sharing Rule to Share the records of Buyer to Sales department with the Access of Read/Write.

Follow the steps from above Activity.

Milestone-10: User Adoption

Activity 1: Create a Record (Farmer)

- Click on App Launcher on left side of screen.
- Search Farm Management & click on it.
- Click on Farmer Tab.
- Click new and fill details & Save.

REPORT

MILESTONE-11/Reports:

Report

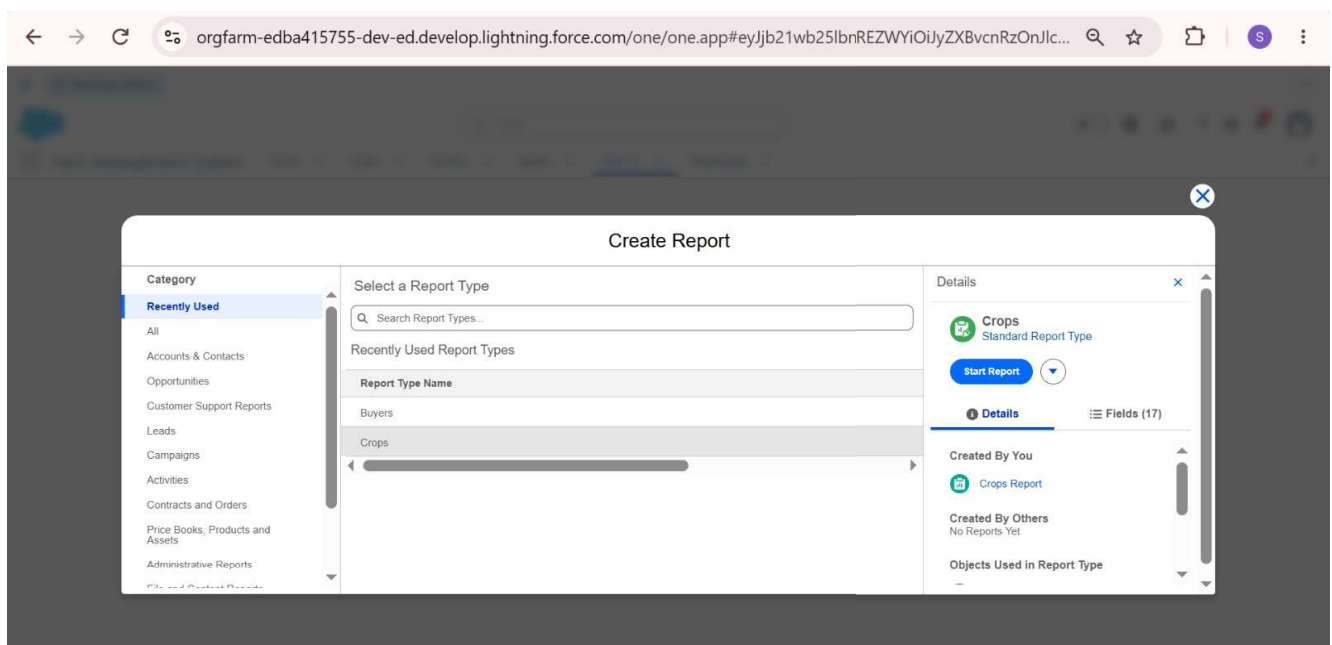
Reports in Salesforce are tools used to collect, organize, and display data stored in objects in a structured format. They help users analyse information and understand business activities by presenting data in tables, summaries, or charts.

Activity 1:

Create a report:

Create a report that displays rating of the account, and which has type and account name.

- 1) Click on app launcher search for reports.
- 2) Click on the new report and select the category has Farmer with Farms.



- 3) In the details section select the option start report.
- 4) In Outline Click on Column Select Which Field you want to display in the Report.
- 5) Click Group, Select Which Fields You Want to Group.
- 6) Save the report by giving label name (Farmer with Farms) and save the folder as a public folder and save the report.

orgfarm-edba415755-dev-ed.develop.lightning.force.com/lightning/r/Report/00Oak00000SCGQfEAP/edit?querySco...

Developer Edition

Farm Management System Farms Crops Farmers Buyers Reports Dashboards

REPORT Farmers with Farms Report Farmers with Farms

Previewing a limited number of records. Run the report to see everything.

Farmer: Farmer Name	size	Soil Quality	Irrigation type
gayathri (1)	12	Composting	Drip Irrigation
Subtotal			
Susivelmurugan (1)	10	Organic	Drip Irrigation
Subtotal			
Total (2)			

Row Counts Detail Rows Subtotals Grand Total

Activity 2:

Create a Report using the Objects Crops, Buyers. Follow the steps from above Activity.

DASHBOARD

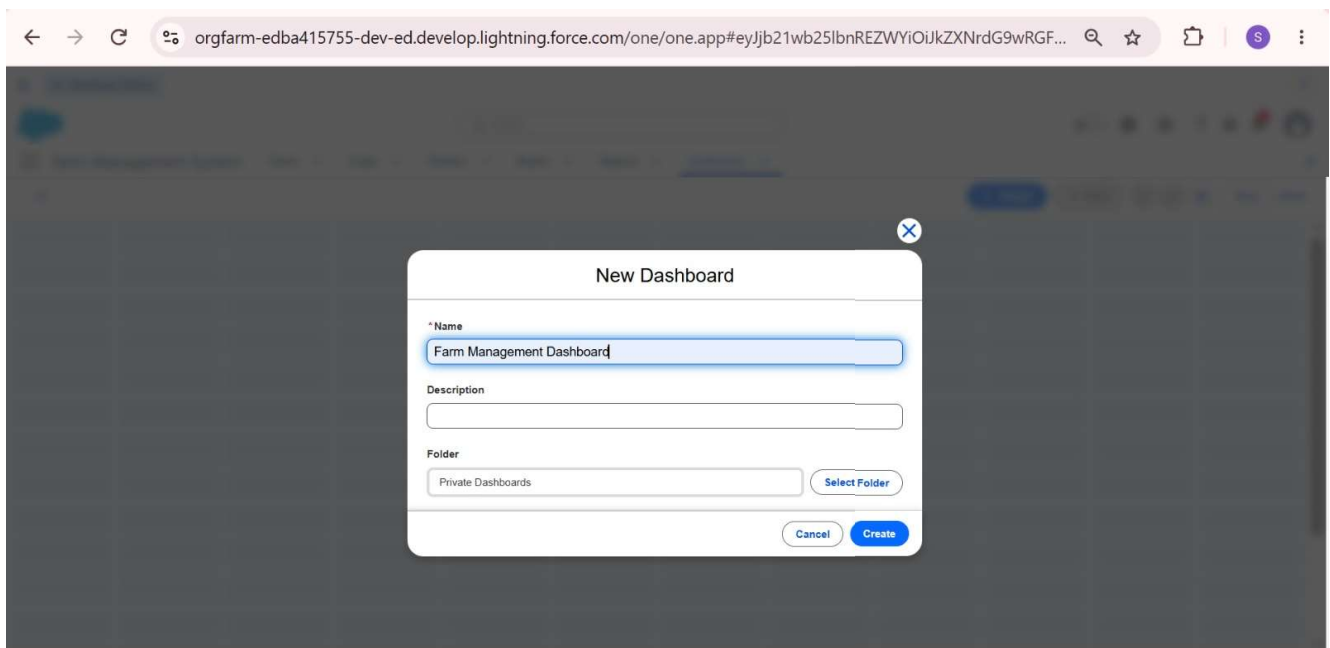
MILESTONE-12/Dashboard:

Dashboards provide more insights than reports as they combine the data from many reports and show a summarized result. Looking at many reports at a time gives the flexibility of combining the results from them quickly. Also summaries in dashboards help us decide on action plans quicker. The dashboards can contain charts, graphs and Tabular data.

Activity 1:

Create a Dashboard:

1. Click the Dashboard tab.
2. Click New Dashboard.
3. Name the dashboard Farmers with farms and click Create.
4. Click +Component.
5. Select the Farmers with Farms and click Select.
6. Select the Gauge chart and click Add.
7. Click Save and then Done.



VIEW AND DASHBOARD

MILESTONE-13/ View Report and Dashboard:

Activity 1: Report

- Click on App Launcher on left side of screen.
- Search Farm Management System & click on it.
- Click on Reports Tab.
- Click on Farmers with Farms & see records

orgfarm-edba415755-dev-ed.develop.lightning.force.com/lightning/o/Report/home?queryScope=mru

Developer Edition

Farm Management System Farms Crops Farmers Buyers Reports Dashboards

Reports
Recent
3 items

REPORTS	Report Name	Description	Folder	Created By	Created On	Subscribed
Recent	Farmers with Farms Report		Private Reports	Susithra V	9/19/2026, 2:15 PM	
Created by Me	Crops Report		Private Reports	Susithra V	9/20/2026, 12:22 AM	
Private Reports	Buyers Report		Private Reports	Susithra V	9/20/2026, 12:23 AM	
Public Reports						
All Reports						
FOLDERS						
All Folders						
Created by Me						
Shared with Me						
FAVORITES						
All Favorites						

Activity 2: Dashboard

- Click on App Launcher on left side of screen.
- Search Farm Management system & click on it.
- Click on Dashboard Tab.
- Click Farmers With farms & see records.

←

→

↺

orgfarm-edba415755-dev-ed.develop.lightning.force.com/lightning/r/Dashboard/01Zak00000bmTtdEAE/view

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Developer Edition

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🔍 Search...

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Farm Management System

Farms

Crops

Farmers

Buyers

Reports

Dashboards

Dashboard

Farm Management Dashboard

As of Sep 20, 2026, 1:50 AM

Viewing as Sushitra V

Refresh

Edit

Subscribe

Farmers with Farms Report

View Report (Farmers with Farms Report)

As of Sep 20, 2026, 1:50 AM

Crops Report

Crop: Crop Name
Rice
maize
wheat

View Report (Crops Report)

As of Sep 20, 2026, 2:04 AM

FUNCTIONAL & PERFORMANCE TESTING

During the testing stage, screenshots were collected from the Salesforce interface to confirm that the system features were functioning properly. The following functionalities were verified:

- Creation of records for Farm, Crop, Farmer, and Buyer
- Validation of required input fields
- Verification of relationships and dependencies between objects
- Testing user access through profiles and roles
- Checking sharing rules and permissions
- Generating reports and dashboards successfully
- Performance testing was conducted to ensure the system responds efficiently and remains stable when multiple users operate the system at the same time.

ADVANTAGES & DISADVANTAGES

Advantages

- Centralized storage of agricultural data
- Minimizes manual paperwork
- Higher data accuracy and reliability
- Easy monitoring of farm activities
- Access to real-time reports and analytics

Disadvantages

- Requires stable internet connectivity
- Dependence on Salesforce platform licensing
- Limited functionality in offline mode

CONCLUSION

The **Farm Management System** (FMS) developed using the Salesforce platform provides an efficient and organized solution for managing agricultural data. The system successfully digitizes important farming information such as farms, crops, farmers, and buyers, and stores it in a centralized cloud-based environment.

By implementing Salesforce features like custom objects, relationships, validation rules, reports, and dashboards, the application ensures accurate data management and easy access to information. The system reduces manual paperwork, minimizes errors, and improves the efficiency of farm management activities.

Through the use of reports and dashboards, users can easily monitor farm operations and analyse agricultural data in real time. This helps farmers and managers make better decisions based on accurate and structured information.

Overall, the Farm Management System demonstrates how modern cloud technology can be effectively used to simplify agricultural management processes. The system improves productivity, enhances data visibility, and supports better planning and decision-making in the agricultural sector.

FUTURE SCOPE

Automation Testing Integration

In the future, functional testing can be automated using tools like Selenium to reduce manual effort and improve testing speed.

AI-Based Testing

Artificial Intelligence can be used to automatically detect bugs, predict failures, and optimize testing strategies.

Cloud-Based Performance Testing

Performance testing can be executed in cloud environments using platforms like Amazon Web Services to simulate real-world traffic from different locations.

Continuous Testing in DevOps

Testing can be integrated with DevOps pipelines using tools such as Jenkins to enable continuous testing during software development.

Improved Security Testing Integration

Future testing frameworks may combine functional, performance, and security testing for more reliable applications.

Real-Time Monitoring and Analytics

Advanced monitoring tools can provide real-time analytics to track application performance and detect issues quickly.

Mobile and Cross-Platform Testing

Future applications will require testing across multiple devices, browsers, and operating systems to ensure compatibility.

Scalability Testing Enhancements

Performance testing will focus more on scalability to handle increasing users and data loads.

Integration with Machine Learning

Machine learning can help automatically generate test cases and identify patterns in system failures.

User Experience (UX) Testing Improvements

Future testing may focus more on user experience to ensure applications are not only functional but also user-friendly.

APPENDIX

Source Code: Salesforce configurations and metadata components

Dataset: Records related to Farm, Crop, Farmer, and Buyer

Tools Used: Salesforce Developer Edition

THANK YOU